



CMPUT 201 A1 (51557)

Fall 2026

PRACTICAL PROG METHODOLOGY

## Territorial Acknowledgement

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The University of Alberta respectfully acknowledges that we are located on Treaty 6 territory, a traditional gathering place for diverse Indigenous peoples, including the Cree, Blackfoot, Métis, Nakota Sioux, Iroquois, Dene, Ojibway/Saulteaux/Anishinaabe, Inuit, and many others whose histories, languages, and cultures continue to influence our vibrant community.

## Instructor Information

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Instructor

Henry Tang

Contact Information

[hktang@ualberta.ca](mailto:hktang@ualberta.ca)

**Office Hours:** By appointment

Lecture Room & Time

VVC 2-215 @9:00am - 9:50am

## Calendar Description

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Introduction to the principles, methods, tools, and practices of the professional programmer. The lectures focus on the fundamental principles of software engineering based on abstract data types and their implementations. The

laboratories offer an intensive apprenticeship to the aspiring software developer. Students use C and software development tools of the Unix environment. Prerequisite: CMPUT 175. Credit cannot be obtained for both CMPUT 201 and CMPUT 275.

## Additional Course Information (General Content)

Course challenging level: **medium to high** | Course workload: **heavy**

## Course Objectives/Outcomes

- Be able to handle any intermediate programming problem by using standard C in linux
- Have the skills to combine your knowledge of program design and data structures (174/175), with useful algorithms and mathematics (272/204/304) and application specific knowledge (291/379) to design and implement non-trivial software
- Understand the design theory of standard C (c99): key programming concepts and their compatibilities

## Required Course Materials



### **C Programming A Modern Approach**

**ISBN:** 9780393979503

**Authors:** Kim N. King

**Publisher:** W.W. Norton

**Publication Date:** 2008

**Edition:** 2

## Course Material Access

[Course Material Access](#) allows students to access their required course materials through Canvas for the duration of the term. A flat fee per term will appear in the

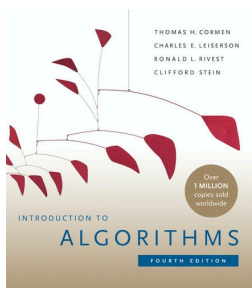
student's Fee Assessment section in Bear Tracks. Digital formats are prioritized, but print materials are provided if digital formats are not available. **This is an automatic opt in program, but students can choose to opt out**, if they decide it is not for them. Some reasons may include:

- the cost of all required course material is less than the program flat fee for the term
- print materials are preferred over digital formats
- access to course material is desired beyond a given term

Students will automatically be opted out if none of their digital or printed course material has a cost associated with it. For more information on how to access your course materials on Canvas, how to opt out, and opt out deadlines, please visit the [Course Material Access](#) website.

## Recommended Readings/Materials

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**Title:** Introduction to Algorithms, fourth edition

**ISBN:** 9780262046305

**Authors:** Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein

**Publisher:** MIT Press

**Publication Date:** 2022-04-05

**Edition:** 4

## On-Line Homework Disclaimer

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Online homework is a component of this course and is provided by a third-party company. Please be aware that this company will be storing assessment information that may be associated with you. As a way to protect your personal information, you may be assigned a random ID to enter into the system. Only the random ID, performance on the online homework, and the affiliation to the University of Alberta for this random ID will be conveyed to the company. You are not required to provide any additional personal information to this company. If you have concerns about this, please contact the instructor of the course.

Students who wish to opt out of the use of the fee-for-service online homework system have the option to...

## Optional Online Learning Resources

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Additional learning resources aimed at facilitating student learning, and perhaps including formative assessment tools, are available from the textbook publisher and may be accessed for a fee paid by the student to the third-party provider (e.g., textbook company). Students choosing to access and use the online resources should note the following:

1. Registration in the system and any monetary transactions are of their own accord and not the responsibility of the University of Alberta.
2. Students should be mindful of protecting personal information and aware of how their personal information might be used and/or shared.
3. Students can consider using a non-identifying email address or account for these purposes.

## Grade Evaluation

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| Title         | Assessment Type | Weight        | Date of Evaluation |
|---------------|-----------------|---------------|--------------------|
| Quizzes       | Quiz            | 2%*9<br>=18%  |                    |
| Lab Exercises | Labwork         | 2%*9<br>=18%  |                    |
| Assignments   | Assignment      | 9%*2<br>=18%  |                    |
| Midterms      | Midterm         | 10%*2<br>=20% |                    |
| Final         | Exam (final)    | 30%           |                    |

For courses with a Final Exam, students must verify the date of the Final Exam on Bear Tracks when the Final Exam Schedule is posted.

Grades are unofficial until approved by the Department and/or Faculty offering the course.

## Re-examination

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There is no possibility of a re-examination in this course.

## Final Letter Grades

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### University of Alberta Grading Policy

Grades reflect judgments of student achievement made by instructors and must correspond to the associated descriptor. These judgements are based on a combination of absolute achievement and relative performance in a class. Faculties may define acceptable grading practices in their disciplines. Such grading practices must align with the University of Alberta [Assessment and Grading Policy](#) and its procedures, which are available online at the [University of Alberta Policies and Procedures Online \(UAPPOL\)](#) website.

Course Grades Obtained by Undergraduate Students

| Descriptor                                                                               | Letter Grade | Grade Point Value |
|------------------------------------------------------------------------------------------|--------------|-------------------|
| Excellent                                                                                | A+           | 4.0               |
| Excellent                                                                                | A            | 4.0               |
| Excellent                                                                                | A-           | 3.7               |
| Good                                                                                     | B+           | 3.3               |
| Good                                                                                     | B            | 3.0               |
| Good                                                                                     | B-           | 2.7               |
| Satisfactory                                                                             | C+           | 2.3               |
| Satisfactory                                                                             | C            | 2.0               |
| Satisfactory                                                                             | C-           | 1.7               |
| Poor                                                                                     | D+           | 1.3               |
| Minimal Pass                                                                             | D            | 1.0               |
| Failure                                                                                  | F or F4      | 0.0               |
| Note: F4 denotes the eligibility of a student to apply for a re-examination of a course. |              |                   |

## Access to Past or Representative Evaluative Material

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1. For each quiz, there is an associated practice quiz;
2. For Midterm #2 and Final exams, there will be some sample written questions. *No full answers are provided for these sample questions.*

## Statement of Expectation for AI Use

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This course fosters the knowledge, critical, and affective skills that are essential to your development as a computing scientist. To achieve the course learning outcomes, you will need to practice and demonstrate these skills without substantive assistance from others or GenAI.

Important: You are welcome to use AI tools as personal learning supports. However AI use is prohibited in assessed work, including pass/fail assignments, discussion boards, graded assignments and exams. Failure to abide by this guideline may be considered an act of cheating and a violation as outlined in the relevant sections of the University of Alberta [Student Academic Integrity Policy](#).

The programs generated by any generative AI tools do not belong to you, and you should not use them as your work for submission, which is considered as plagiarism. This applies to all LLMs and other Generative Models such as ChatGPT, Lex, Page, DALL-E, Google BARD/Gemini, Microsoft Bing/Copilot, DeepSeek, Qwen, Llama, and others.

If you use any of them to assist your work **you must cite it**. This includes the corporation that made the AI, the AI, Subject, and Date. You are expected to review, customize, fix, and understand all code produced by them. Your marks will be reduced (potentially to zero) if you cannot explain the code you submitted, or further subject to academic integrity investigation.

In short, you will be graded on your ability to understand your submitted work, and it is strongly recommended that you **do not use any such tools** for this course. As an example:

// The following function has code or ideas from Microsoft Copilot "How do I write a merge sort in C?"

// Prompted by Henry Tang on 2026-07-23

```
void mergeSort(int [] arr, int n) {
```

// The following function is adapted from:

<https://stackoverflow.com/questions/12674094/array-to-pointer-decay-and-passing-multidimensional-arrays-to-functions#answer-12674201>

// Answer by KerrekSB

// Taken by Henry Tang on 2026-09-02

```
char (*p)[8] = moreThings;
```

## Missed Term Work/Final Exam Due to Non-medical Protected Grounds (e.g., religious beliefs)

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When a term assessment or final exam presents a conflict based on [non-medical protected grounds](#), students can register with Accessibility and Accommodations via their [Register for Accommodations](#) website. Students can review their eligibility and choose the registration process specific for **Accommodations Based on Non-medical Protected Grounds**.

It is imperative that students review the dates of all course assessments upon receipt of the course syllabus, and register **AS SOON AS POSSIBLE** to ensure the timely application of the accommodation. Students who register later in the term may experience unavoidable delays in the processing of the registration, which can affect the accommodation.

## Policies for Late and Missed Term Work

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### Late policies:

No late submissions are accepted.

**Missed Term Work/Final Exam Due to Non-medical Protected Grounds (e.g., religious beliefs):**

When a term assessment or final exam presents a conflict based on [non-medical protected grounds](#), students can register with the Academic Success Centre for accommodations via their [Register for Accommodations](#) website. Students can review their eligibility and choose the registration process specific for Accommodations Based on Non-medical Protected Grounds.

It is imperative that students review the dates of all course assessments upon receipt of the course syllabus, and register **AS SOON AS POSSIBLE** to ensure the timely application of the accommodation. Students who register later in the term may experience unavoidable delays in the processing of the application, which can affect the accommodation.

### **Excused Absences:**

A student who cannot complete a term work, except the final exam, due to incapacitating illness, severe domestic affliction or other compelling reasons must contact the instructor within 7 calendar days of missing the assessment, or as soon as possible, to request an excused absence. If an excused absence is granted, then the percentage of the term work is moved to the final exam. An excused absence is a privilege and not a right. There is no guarantee that an absence will be excused. Misrepresentation of facts to gain an excused absence is a serious breach of the Student Academic Integrity Policy. In all cases, instructors may request adequate documentation to substantiate the reason for the absence at their discretion.

## Re-evaluation of Term Work

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Re-evaluation requests should be made to the marker(s) within 7 calendar days of the marking (re-marking could reduce your mark); after that time, marks cannot be changed.

## Deferred Final Examination

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A student who cannot write the final examination due to incapacitating illness, severe domestic affliction, or other compelling reasons can apply for a deferred final examination. Such an application must be made to the student's home Faculty Office within two working days of the missed exam and must be supported by appropriate documentation or a Statutory Declaration (see calendar on [Attendance](#)). Deferred examinations are a privilege and not a right; there is no

guarantee that a deferred examination will be granted. The Faculty may deny deferral requests in cases where less than 50% of term work has been completed. Misrepresentation of facts to gain a deferred examination is a serious breach of the [Student Academic Integrity Policy](#).

## Academic Integrity and Student Conduct

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The University of Alberta is committed to the highest standards of academic integrity and honesty. Students are expected to be familiar with these standards regarding academic honesty and to uphold the policies of the University in this respect. Students are particularly urged to familiarize themselves with the provisions of the [Student Academic Integrity Policy](#) (on the [University of Alberta Policies and Procedures Online \(UAPPOL\) website](#)) and avoid any behaviour which could potentially result in suspicions of cheating, plagiarism, misrepresentation, and/or unauthorised collaboration. Academic dishonesty is a serious offence and can result in suspension or expulsion from the University.

The University of Alberta is also committed to maintaining a learning environment that fosters the safety, security, and the inherent dignity of each member of the community, ensuring students conduct themselves accordingly by avoiding behaviour stipulated in the [Student Conduct Policy](#) (e.g., discrimination, harassment, physical assault).

All students are expected to consult the [Academic Integrity website](#) for clarification on the various academic offences. All forms of academic dishonesty are unacceptable at the University. Unfamiliarity of the rules, procrastination or personal pressures are not acceptable excuses for committing an offence. Listen to your instructor, be a good person, ask for help when you need it, and do your own work -- this will lead you toward a path to success. Any academic integrity concern in this course will be reported to the College of Natural and Applied Sciences. Suspected cases of non-academic misconduct will be reported to the Office of Student Success and Experience (formerly the Office of the Dean of Students). The College, Faculty, and Dean of Students are committed to student rights and responsibilities, and adhere to due process and administrative fairness, as outlined in the [Student Academic Integrity Policy](#) and the [Student Conduct](#)

[Policy](#). Please refer to the policies for details on inappropriate behaviours and possible sanctions.

The College of Natural and Applied Sciences (CNAS) has created an [Academic Integrity for CNAS Students](#) website. Website content includes the importance of academic integrity, examples of academic misconduct & possible sanctions, and the academic misconduct & appeal process. Students can also access this material as an [online, self-directed Canvas course](#) and complete assessments to test their knowledge.

The Office of Student Conduct and Accountability has also made the following Canvas course available for all students: [Academic Citizenship: Integrity and Belonging in the Learning Community](#).

*"Integrity is doing the right thing, even when no one is watching" -- C.S. Lewis*

## Contract Cheating and Misuse of University Academic Materials or Other Assets

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Contract cheating describes the form of academic dishonesty where students get academic work completed on their behalf, which they submit for academic credit as if they had created it themselves. Contract cheating may or may not involve the payment of a fee to a third party, who then creates the work for the student.

### **Examples include:**

1. Getting someone to write an essay or research paper for you.
2. Getting someone to complete your assignment or exam for you.
3. Posting an essay, assignment, or exam question to a tutorial or study website; the question is answered by a "content expert", then you copy it and submit it as your own answer.
4. Posting your solutions to a tutorial/study website, public server, or group chat and/or copying solutions that were posted to a tutorial/study website, public server, or group chat.
5. Sharing your login credentials to the course management system (e.g., Canvas) and allowing someone else to complete your assignment or exam remotely.
6. Using an artificial intelligence bot or text generator tool to complete your essay, research paper, assignment, or exam solutions for you (without the

- instructor's permission).
7. Using an online grammar checker to "fix" your essay, research paper, assignment, or exam solutions for you (without the instructor's permission).

Contract cheating companies thrive on making students believe that they cannot succeed without their help; they attempt to convince students that cheating is the only way to succeed.

Uploading the instructor's teaching materials (e.g., course outlines, lecture slides, assignment, or exam questions, etc.) to tutorial, study or note-sharing websites, public servers, or chat apps is a copyright infringement and constitutes the misuse of University academic materials or other assets. Receiving assignment solutions or answers to exam questions from an unauthorized source puts you at risk of receiving inaccurate information.

## Appropriate Collaboration

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Students need to be able to recognize when they have crossed the line between appropriate collaboration and inappropriate collaboration. If students are unsure, they need to ask instructors to clarify what is allowed and what is not allowed.

### **Here are some tips to avoid copying on assessments:**

1. Do not write down something that you cannot explain to your instructor.
2. When you are helping other students, avoid showing them your work directly. Instead, explain your solution verbally. Allowing your work to be copied is also considered inappropriate collaboration.
3. It is also possible that verbally discussing the solution in too much detail may result in written responses that are too similar. Try to keep discussions at a general or higher level.
4. If you find yourself reading another student's solution, do not write anything down. Once you understand how to solve the problem, remove the other person's work from your sight and then write up the solution to the question yourself. Looking back and forth between someone else's paper and your own paper is almost certainly copying and considered inappropriate collaboration.
5. If the instructor or TA writes down part of a solution in order to help explain it to you or the class, you cannot copy it and hand it in for credit. Treat it the same way you would treat another student's work with respect to copying, that is, remove the explanation from your sight and then write up the solution yourself.

6. There is often more than one way to solve a problem. Choose the method that makes the most sense to you rather than the method that other students happen to use. If none of the ideas in your solution are your own, there is a good chance it will be flagged as copying.

**GitHub Use Policy:**

CMPUT 201 has its own GitHub organization on GitHub for each term. By the end of the term, you will have multiple private repositories in this organization: one for each of your labs and/or assignments. All your repos will automatically be individual and private.

When demoing your lab exercise, you must pull your code from that lab's GitHub repo where the commit before the deadline is. Thus, you should make sure your final code is pushed to your GitHub lab repo. Make sure that you put each lab exercise code in the correct repo. When marking labs and assignments, your TA will pull the code from your GitHub repo at the time of the deadline. This is the version that will be marked.

You should not make another GitHub repository to publicly share CMPUT 201 code, which is a breach of the [Student Academic Integrity Policy](#), even after you complete the course. The repos you create in this course will not be deleted. You will always have access to these repos; however, they must remain private and you cannot share them.

## Exam Conduct

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Please refer to the [Examinations](#) section of the Academic Calendar for more details on Conduct of Exams.

Some key points to be aware of:

- Your student photo ID is required at exams to verify your identity.
- Students must arrive at the specified time to take the exam. Once the exam has started, students must remain in the physical in-person or remote environment for at least 30 minutes. Students who arrive more than 30 minutes late for an in-person exam will not be permitted to take the exam. Students who arrive more than 30 minutes late for an online exam may have their exam attempt removed or disqualified by the instructor. In both cases, students may apply for a deferred examination.
- All cell phones must be turned off and stored in your bags.

# University Policy

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## Withdrawals

See the [University Calendar](#) for the relevant [add/drop deadlines](#) for each term.

## Course Outlines

Policy about course outlines can be found in the [Academic Regulations, Evaluation Procedures and Grading section](#) of the University Calendar.

## Student Academic Integrity

The University of Alberta is committed to the highest standards of academic integrity and honesty. Students are expected to be familiar with these standards regarding academic honesty and to uphold the policies of the University in this respect. Students are particularly urged to familiarize themselves with the provisions of the Student Academic Integrity Policy and the Student Conduct Policy (on the [University of Alberta Policies and Procedures Online](#) (UAPPOL) website) and avoid any behaviour which could potentially result in suspicions of cheating, plagiarism, misrepresentation of facts and/or participation in an offence. Academic dishonesty is a serious offence and can result in suspension or expulsion from the University.

## Recordings

Audio or video recording, digital or otherwise, of lectures, labs, seminars or any other teaching environment by students is allowed only with the prior written consent of the instructor or as a part of an approved accommodation plan. Student or instructor content, digital or otherwise, created and/or used within the context of the course is to be used solely for personal study, and is not to be used or distributed for any other purpose without prior written consent from the content authors.

## Accommodations for Students

In accordance with the University of Alberta's [Accommodation Policy](#) and [Discrimination and Harassment Policy](#), accommodation support is available to eligible students who encounter limitations or restrictions to their ability to perform the daily activities necessary to pursue studies at a post-secondary level due to medical conditions and/or non-medical protected grounds. Accommodations are coordinated through the [Accessibility and Accommodations service](#), and students can learn more about eligibility on the [Register for Accommodations website](#).

It is recommended that students apply AS SOON AS POSSIBLE in order to ensure sufficient time to complete accommodation registration and coordination. Students are advised to review and adhere to published deadlines for accommodation approval and for specific accommodation requests (e.g., exam registration submission deadlines). Students who request accommodations less than a month in advance of the academic term for which they require accommodations may experience unavoidable delays or consequences in their academic programs, and may need to consider alternative academic schedules.

## Student Supports

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**Faculty of Science Student Services:** The [Faculty of Science Student Services](#) office is located on the main floor of the Centennial Centre for Interdisciplinary Sciences (CCIS). This office can assist with the planning of [Your Academics](#), and provide information related to [Student Life & Engagement](#), [Internship and Careers](#), and [Study Abroad](#) opportunities. Please visit [Advising](#) for more information about what Faculty Academic Advisors can assist you with.

**Student Services Directory:** The [Student Services Directory](#) helps U of A students find and access support quickly and effectively, bringing together a catalogue of student services from across the university in one convenient location.

**Writing and Learning Centre:** Get support when and where you need it. The centre, located in 2-10 Cameron Library, offers both in-person and virtual supports for undergraduate and graduate students. Strengthen your academic

skills by attending a workshop or booking a one-on-one appointment. Their [website](#) provides more information about drop-in appointments, writing groups, Canvas modules and special programming to find what works best for you. Workshops and appointments are free for current U of A students.

**Feeling Stressed, Anxious, or Upset?** It's normal for us to have different mental health experiences throughout the year. Know that there are people who want to help. You can reach out to your friends and access a variety of supports available on and off campus at the [Need Help Now](#) webpage or by calling the 24-hour Distress Line: 780-482-4357 (HELP).

## Learning and Working Environment

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The Faculty of Science is committed to ensuring that all students, faculty and staff are able to work and study in an environment that is safe and free from discrimination, harassment, and violence of any kind. It does not tolerate behaviour that undermines that environment. This includes virtual environments and platforms.

If you are experiencing harassment, discrimination, fraud, theft or any other issue and would like to get confidential advice, please contact any of these campus services:

- [Office of Safe Disclosure & Human Rights](#): A safe, neutral and confidential space to disclose concerns about how the University of Alberta policies, procedures or ethical standards are being applied. They provide strategic advice and referral on matters such as discrimination, harassment, duty to accommodate and wrong-doings. Disclosures can be made in person or online using the [Online Reporting Tool](#).
- [University of Alberta Protective Services](#): Peace officers dedicated to ensuring the safety and security of U of A campuses and community. Staff or students can contact UAPS to make a report if they feel unsafe, threatened, or targeted on campus or by another member of the university community.
- [Office of the Student Ombuds](#): A confidential and free service that strives to ensure that university processes related to students operate as fairly as possible. They offer information, advice, and support to students, faculty, and

staff as they deal with academic, discipline, interpersonal, and financial issues related to student programs.

- [Office of Student Success and Experience](#): They can assist students in navigating services to ensure they receive appropriate and timely resources. For students who are unsure of the support they may need, are concerned about how to access services on campus, or feel like they may need interim support while they wait to access a service, this office is there to help.

## Disclaimer

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Any typographical errors in this syllabus are subject to change and will be announced in class and/or posted on the course website. The date of final examinations is set by the Registrar and takes precedence over the final examination date reported in the syllabus.

## Copyright

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Mr. Henry Tang, Department of Computing Science, Faculty of Science, University of Alberta (2026).

Dr. Guohui Lin and Mr. Henry Tang, Department of Computing Science, Faculty of Science, University of Alberta (2025-2026).